FNCDP 2025 Python Toolkit

National Cybersecurity & Data Protection Framework Integration Suite

This project is a modular Python toolkit designed to digitize, analyze, and integrate the ACN Cybersecurity Framework (FNCDP 2025 v2.1). It transforms the official static Excel file into a complete software suite, helping security analysts, auditors, and IT teams manage gap analysis workflows and integrate security controls into enterprise environments.

System Architecture (6 Key Modules)

01 Core & Database

File: `fncdp_core.py`
Ingestion engine. Reads ACN Excel, reconstructs the hierarchy (Functions → Categories → Subcategories), and normalizes data into SQLite (`fncdp_2025.db`).

02 Maturity Assessment

File: `1_assessment.py`
Manages gap analysis. Generates Excel templates for control Owners, reads scores (0-5), and computes average maturity per Category and Function.

03 CLI Search Engine

File: `2_ricerca.py`
Command-line tool for auditors. Search controls by keyword (e.g. backup) or standard (GDPR, ISO 27001, NIST) and export results to Excel.

04 REST API Backend

File: `3_api_server.py`
FastAPI & Uvicorn backend. Exposes JSON HTTP endpoints for third-party software (ServiceNow, Jira, SIEMs). Includes Swagger UI at `/docs`.

05 CISO Dashboard Planned

File: `dashboard.py`
Web UI for decision-makers. Visualizes metrics, KPIs, category distributions, dynamic filters, and control matrices in real time.

06 Doc Generator Planned

File: `genera_pdf.py`
ReportLab-based script to automatically generate single-page PDF README reports styled in technical English for CSIRT ACN.

Strategic Development Roadmap

- **Compliance Integration:** Native support and mapping for **NIS2** Directive and **DORA** Regulation requirements.
- **Security Upgrade:** Implementation of **JWT (JSON Web Tokens)** authentication to secure REST API endpoints.
- **Enterprise Database:** Migration path from SQLite to **PostgreSQL** to support multi-user environments and high concurrency.